

BEE 4750 Homework 1: Introduction to Using Julia

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Due Date

Thursday, 9/11/25, 9:00pm

Overview

Instructions

- Problem 1 consist of a series of code snippets for you to interpret and debug. You will be asked to identify relevant error(s) and fix the code.
- Problem 2 asks you to write code to implement several simple functions using more general programming and some which are unique to Julia (which might mean that you need to look at the documentation or use Julia's help function, which is accessed with `?`, e.g. `?sum` for help with the `sum()` function) or to analyze Julia syntax.
- Problem 3 asks you to convert a verbal description of a wastewater treatment system into a Julia function, and then to use that function to explore the impact of different wastewater allocation strategies.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
Pkg.add("DataFrames")
```

```
Activating project at `~/BEE 4750 /HW 1/bee4750-hw1-rv297`
  Updating registry at `~/julia/registries/General.toml`
  Resolving package versions...
  Installed InlineStrings          v1.4.5
  Installed DataValueInterfaces    v1.0.0
  Installed PooledArrays           v1.4.3
  Installed SentinelArrays         v1.4.8
  Installed InvertedIndices        v1.3.1
  Installed TableTraits            v1.0.1
  Installed PrettyTables           v2.4.0
  Installed DataFrames             v1.7.1
  Installed Crayons                v4.1.1
  Installed Tables                 v1.12.1
  Installed IteratorInterfaceExtensions v1.0.0
  Installed StringManipulation     v0.4.1
  Updating `~/BEE 4750 /HW 1/bee4750-hw1-rv297/Project.toml`
[a93c6f00] + DataFrames v1.7.1
  Updating `~/BEE 4750 /HW 1/bee4750-hw1-rv297/Manifest.toml`
[a8cc5b0e] + Crayons v4.1.1
[a93c6f00] + DataFrames v1.7.1
[e2d170a0] + DataValueInterfaces v1.0.0
[842dd82b] + InlineStrings v1.4.5
[41ab1584] + InvertedIndices v1.3.1
[82899510] + IteratorInterfaceExtensions v1.0.0
[2dfb63ee] + PooledArrays v1.4.3
[08abe8d2] + PrettyTables v2.4.0
[91c51154] + SentinelArrays v1.4.8
[892a3eda] + StringManipulation v0.4.1
[3783bdb8] + TableTraits v1.0.1
[bd369af6] + Tables v1.12.1
[9fa8497b] + Future v1.11.0
  Info Packages marked with `~` have new versions available but compatibility constraints
Precompiling project...
 1041.8 ms  IteratorInterfaceExtensions
   940.1 ms  DataValueInterfaces
   938.4 ms  InvertedIndices
   999.3 ms  PooledArrays
```

```

1003.5 ms    InlineStrings
1442.7 ms    Crayons
1567.0 ms    SentinelArrays
688.8 ms     TableTraits
809.3 ms     InlineStrings → ParsersExt
1283.8 ms    StringManipulation
882.8 ms     Tables
10402.0 ms   PrettyTables
22193.2 ms   DataFrames
1320.6 ms    Latexify → DataFramesExt
14 dependencies successfully precompiled in 37 seconds. 241 already precompiled.

```

Standard Julia practice is to load all needed packages at the top of a file. If you need to load any additional packages in any assignments beyond those which are loaded by default, feel free to add a `using` statement, though [you may need to install the package](#).

```

using Random
using Plots
using GraphRecipes
using LaTeXStrings
using Distributions

```

```

# this sets a random seed, which ensures reproducibility of random number generation. You should
Random.seed!(1)

```

`TaskLocalRNG()`

Problems (Total: 30 Points)

Problem 1 (12 points)

The following subproblems all involve code snippets that require debugging. You are encouraged to use online resources (*e.g.* Julia documentation and forums, Stack Overflow, Reddit, etc) to help find diagnose error messages and find solutions, but make sure that you clearly document which resources you used and how you used them.

Problem 1.1

You've been tasked with writing code to identify the minimum value in an array. You cannot use a predefined function. Your colleague suggested the function below, but it does not return the minimum value.

```

function minimum(array)
    # initialize the minimum value counter
    min_value = array[1]
    # update minimum values
    for i in 1:length(array)
        if array[i] < min_value
            min_value = array[i]
        end
    end
    # return found minimum
    return min_value
end

array_values = [89, 90, 95, 100, 100, 78, 99, 98, 100, 95]
@show minimum(array_values);

```

```
minimum(array_values) = 78
```

What is the logical flaw in the code? Fix it and show that your fixed code solves the problem.

1.1 Solution:

The flaw in the original code was initializing `min_value` to 0. This can incorrectly return 0 if, for example, the array only contains positive numbers because there would never be a value smaller than 0.

To fix this issue I initialized `min_value` to the first element of the array (`array[1]`). This way the starting value is always valid and within the array. The loop then correctly updates `min_value` as it goes through each element in the array and returns the true minimum value, as can be seen above when 78 is the given minimum of the given array. ##### Problem 1.2

Your team is trying to compute the average grade for your class, but the following code produces an `UndefVarError`.

```

# enter student grade vector
student_grades = [89, 90, 95, 100, 100, 78, 99, 98, 100, 95]
# compute class average
function class_average(grades)
    average_grade = mean(student_grades)
    return average_grade
end
average_grade = class_average(student_grades)
@show average_grade;

```

```
average_grade = 94.4
```

What does this `UndefVarError` mean? Why does this code produce one? Fix the error and solve the problem.

1.2 Solution:

According to Stack Overflow ‘`UndefVarError`’ has to do with the scope of variables. In the code above, `average_grade` only exists in the function `class_average`. Calling it in the global/main scope will not work since it does not exist there. To fix this, we can call the correct average value in the global scope using the function `class_average` and store it in the variable `average_grade`. Then, Julia will correctly show the average grade as 94.4 with no errors!

Problem 1.3

Your team wants to know the expected payout of an old Italian dice game called *passadieci* (which was analyzed by Galileo as one of the first examples of a rigorous study of probability). The goal of *passadieci* is to get at least an 11 from rolling three fair, six-sided dice. Your strategy is to compute the average wins from 1,000 trials, but the code you’ve written below produces a `MethodError`.

```
# function to simulate a passadieci roll: 3 6-sided dice
function passadieci()
    # this rand() call samples 3 values from the vector [1, 6]
    roll = rand(1:6, 3)
    return roll
end
# set number of trials and initialize outcome vector
n_trials = 1_000
outcomes = zeros{Int, n_trials}
# simulate number of passadieci rolls and count wins
for i = 1:n_trials
    outcomes[i] = (sum(passadieci()) > 11)
end
win_prob = sum(outcomes) / n_trials # compute average number of wins
@show win_prob;
```

```
win_prob = 0.382
```

What does this `MethodError` mean? Why does this code produce one? Fix the error and solve the problem.

1.3 Solution:

I was not able to find much on StackOverflow or Julia documentation/forums so I asked ChatGPT what ‘MethodError’ means. It stated that Julia could not find any function definition/method that matches the arguments given. So I realized that in this case, `zeros(n_trials)` returned the integer 0 but the code requires for an array when calling `outcomes[i]`, hence the ‘MethodError’. To fix this, I made `outcomes = zeros{Int, n_trials}` so that `outcomes` is now an integer array of length `n_trials`. At this point, `outcomes[i]` can be called properly.

Problem 1.4

You’re interested in writing some code to remove the mean of a vector from all of its components. You’ve written the following code and tried to test it on a random vector, but your code returns a `MethodError`.

```
# function to remove mean from a vector
function remove_mean(vect)
    # fucntion to compute the mean
    function compute_mean(vect)
        element_sum = 0 # initialize sum
        # compute mean and return
        for v in vect
            element_sum += v
        end
        return element_sum / length(vect)
    end

    m = compute_mean(vect) # compute mean
    # return demeaned vector

    new_vect = copy(vect)
    for i in 1:length(vect)
        new_vect[i] = vect[i] - m
    end
    return new_vect
end

random_vect = rand(1_000)
@show remove_mean(random_vect)
```

```
remove_mean(random_vect) = [0.41919142411294885, -0.04477120169153992, -0.15987063495003362,
```

```
1000-element Vector{Float64}:
 0.41919142411294885
```

```

-0.04477120169153992
-0.15987063495003362
-0.3165390583665356
-0.04345183269168251
 0.4600428661267356
-0.4814808404373695
 0.05265958609672772
-0.019033198532207418
 0.11752715148103854

 0.1701029484937252
-0.45332724205642416
-0.4743008238822817
-0.4129715382326393
 0.23830378602543056
 0.06271140871181402
 0.11506229654060751
-0.1638906511382653
-0.0897920909169232

```

What does this `MethodError` mean? Why does this code produce one? Fix the error and solve the problem.

Problem 1.4 solution:

As stated in problem 1.3, ‘`MethodError`’ means that Julia could not find any function definition/method that matches the arguments given. In this case, the error is thrown because `vect - m` is trying to subtract a scalar from a vector, which is not possible. To fix this, I made a new vector, `new_vect`, to modify and subtract the mean from. I did this by copying `vect` with the method `copy()`, which I found on Julia documentation. Then I used a for loop to index through and subtract `m` from each element in `new_vect`, which now works because it is a scalar subtracted from a scalar. Finally, `new_vect` returns the correct vector we are looking for.

Problem 2 (18 points)

Cheap Plastic Products, Inc. is operating a plant that produces $100\text{m}^3/\text{day}$ of wastewater that is discharged into Pristine Brook. The wastewater contains $1\text{kg}/\text{m}^3$ of YUK, a toxic substance. The US Environmental Protection Agency has imposed an effluent standard on the plant prohibiting discharge of more than $20\text{kg}/\text{day}$ of YUK into Pristine Brook.

Cheap Plastic Products has analyzed two methods for reducing its discharges of YUK. Method 1 is land disposal, which costs $X_1^2/20$ dollars per day, where X_1 is the amount of wastewater

disposed of on the land (m^3/day). With this method, 20% of the YUK applied to the land will eventually drain into the stream (*i.e.*, 80% of the YUK is removed by the soil).

Method 2 is a chemical treatment procedure which costs \$1.50 per m^3 of wastewater treated. The chemical treatment has an efficiency of $e = 1 - 0.005X_2$, where X_2 is the quantity of wastewater (m^3/day) treated. For example, if $X_2 = 50\text{m}^3/\text{day}$, then $e = 1 - 0.005(50) = 0.75$, so that 75% of the YUK is removed.

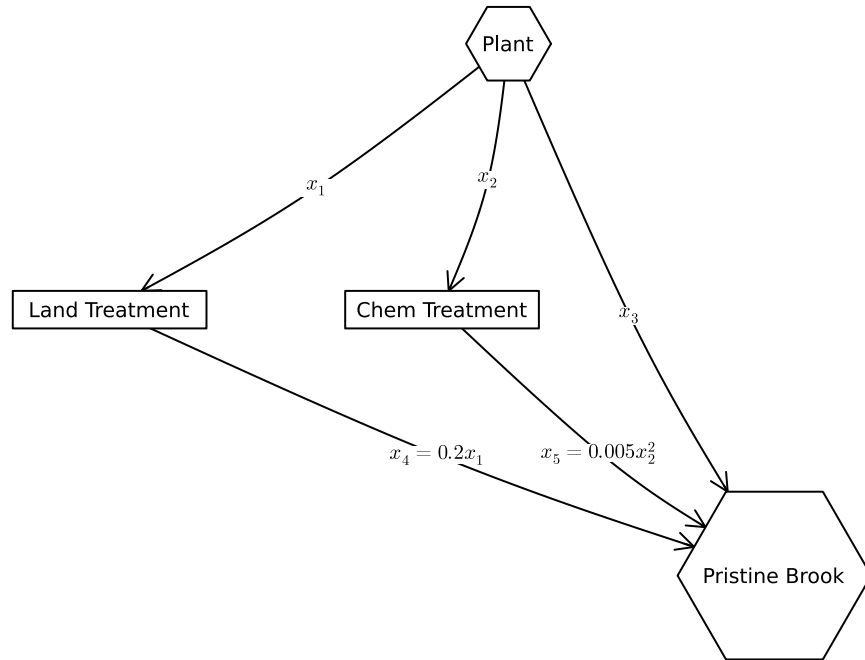
Cheap Plastic Products is wondering how to allocate their wastewater between these three disposal and treatment methods (land disposal, chemical treatment, and direct disposal) to meet the effluent standard while keeping costs manageable.

The flow of wastewater through this treatment system is shown in Figure 1. Modify the edge labels (by editing the `edge_labels` dictionary in the code producing Figure 1) to show how the wastewater allocations result in the final YUK discharge into Pristine Brook. For the `edge_label` dictionary, the tuple (i, j) corresponds to the arrow going from node i to node j . The syntax for any entry is $(i, j) \Rightarrow \text{"label text"}$, and the label text can include mathematical notation if the string is prefaced with an L, as in `L"x_1"` will produce x_1 .

```
using Random
using Plots
using GraphRecipes
using LaTeXStrings
using Distributions
A = [0 1 1 1;
      0 0 0 1;
      0 0 0 1;
      0 0 0 0]

names = ["Plant", "Land Treatment", "Chem Treatment", "Pristine Brook"]
# modify this dictionary to add labels
edge_labels = Dict{Tuple{Int, Int}, String}((1, 2) => L"x_1", (1, 3) => L"x_2", (1, 4) => L"x_3", (2, 4) => L"x_4 = 0.2",
(3, 4) => L"x_4 = 0.2")
shapes=[:hexagon, :rect, :rect, :hexagon]
xpos = [0, -1.5, -0.25, 1]
ypos = [1, 0, 0, -1]

p = graphplot(A, names=names, edgelabel=edge_labels, markersize=0.15, markershapes=shapes, margin=:auto)
display(p)
```

Explanation of Figure 1

This image is showing the concentrations of YUK from when they leave the Plant until they reach Pristine Brook. The concentrations that leave the plant are : x_1 , x_2 , and x_3 . The concentrations that reach Pristine Brook include: x_4 , x_5 , and x_3 . x_1 and x_2 change after going through treatment, as shown above.

Problem 2.1

Formulate a mathematical model for the treatment cost and the amount of YUK that will be discharged into Pristine Brook based on the wastewater allocations. This is best done with some equations and supporting text explaining the derivation. Make sure you include, as additional equations in the model, any needed constraints on relevant values. You can find some basics on writing mathematical equations using the LaTeX typesetting syntax [here](#), and a cheatsheet with LaTeX commands can be found on the course website's [Resources page](#).

Solution:

The wastewater contains $1\text{kg}/\text{m}^3$ so when $x_1 \text{ m}^3/\text{day}$ leaves the plant and heads toward the land treatment the quantity of YUK is also $x_1 \text{ kg/day}$. Once this wastewater has passed the land treatment, 80% of the YUK is removed by the soil so $x_4 = 0.2x_1 \text{ kg/day}$. Similarly, the x_2

wastewater flow has x_2 kg/day of YUK. After it passes the chemical treatment with efficiency $e = 1 - 0.005 x_2$, it will have a new concentration of YUK $x_5 = (1-e) x_2 = 0.005 x_2^2$. Finally the third stream, x_3 , leaving the plant does not go through any treatment so it will have the same concentration of x_3 kg/day of YUK. The final amount of YUK that will be discharged into the Pristine Brook will be $0.2 x_1 + 0.005 x_2^2 + x_3$ 20 kg/day with two constraints: $x_1 + x_2 + x_3 = 100 \text{ m}^3/\text{day}$ and the final amount of YUK must be 20 kg/day.

The treatment cost of the water treatment is $x_1^2 / 20 + \$1.50 x_2$, which includes the cost of land and chemical treatment.

In summary, these are the equations and constraints to model the YUK concentrations going into Pristine Brook and treatment costs:

$$\text{Total YUK concentrations} = 0.2 x_1 + 0.005 x_2^2 + x_3$$

$$\text{Total treatment cost} = x_1^2 / 20 + \$1.50 x_2$$

Constraints:

1. $x_1 + x_2 + x_3 = 100 \text{ m}^3/\text{day}$
2. Total YUK concentrations must be 20 kg/day

Problem 2.2

Implement your systems model as a Julia function which computes the resulting YUK discharge and cost for a particular treatment plan. You can return multiple values from a function with a [tuple](#), as in:

```
function multiple_return_values(x, y)
    return (x+y, x*y)
end

a, b = multiple_return_values(2, 5)
@show a;
@show b;
```

```
a = 7
b = 10
```

To evaluate the function over vectors of inputs, you can *broadcast* the function by adding a decimal `.` before the function arguments and accessing the resulting values by writing a *comprehension* to loop over the individual outputs in the vector:

```
x = [1, 2, 3, 4, 5]
y = [6, 7, 8, 9, 10]
```

```
output = multiple_return_values.(x, y)
a = [out[1] for out in output]
b = [out[2] for out in output]
@show a;
@show b;
```

```
a = [7, 9, 11, 13, 15]
b = [6, 14, 24, 36, 50]
```

```
function yuk_output(x1, x2, Q = 100)

    x3 = 100 - x1 - x2 #find x3 - water receiving NO treatment

    discharge = 0.2 * x1 + 0.005 * x2^2 + x3 #total YUK concentration going into Pristine Br
    cost = ( x1^2 / 20 ) + 1.50 * x2 #total cost of the water treatment for x1 and x2 ($/day)

    #Discharge of YUK into the Pristine Brok must be less than 20 kg/day, this part of the c
    if discharge <= 20 #if YUK concentration is no more than 20kg/day, then standard is met
        meets_standard = true
    else
        meets_standard = false #if it is more than 20kg/day, standard is NOT met
    end
    return(discharge, cost, meets_standard)

end

"""
x1 = 20
x2 = 30
(d, c, m) = yuk_output(x1, x2, 100)
@show d;
@show c;
@show m;
"""

# Problem 2.3
using Random, Distributions, Plots
Random.seed!(42)
```

```

Q = 100 # max flow leaving plant
N = 1000 # N of trials

# set up the random trials
X = rand(Dirichlet(3,1.0), N)
x1 = Q .* X[1, :]
x2 = Q .* X[2, :]

d = similar(x1) # discharge
c = similar(x2) # cost
s = falses(N) # whether discharge into Pristine Brook meets the standard

for i in 1:N
    d[i], c[i], s[i] = yuk_output(x1[i], x2[i], Q) # test the model for each of the random trials
end

# Find all the tests that do NOT exceed 20 kg/day of YUK into Pristine Brook
using DataFrames
feasible_trials = DataFrame( x1 = x1[s], x2 = x2[s], x3 = 100 .- x1[s] .- x2[s], discharge = s)
print(feasible_trials)

# Scatter plot of all trials
scatter(d[.!s], c[.!s], label="Exceed 20 kg/day limit", ms=4, markerstrokewidth=0,
        xlabel="YUK (kg/day)", ylabel="Cost (\$/day)") # plot all points that are above the limit
scatter!(d[s], c[s], label="Meet 20 kg/day limit", ms=4, markerstrokewidth=0) # plot all points that meet the limit
vline!([20], linestyle=:dash, color=:red, label="YUK limit") # mark the 20 kg/day limit with a vertical line

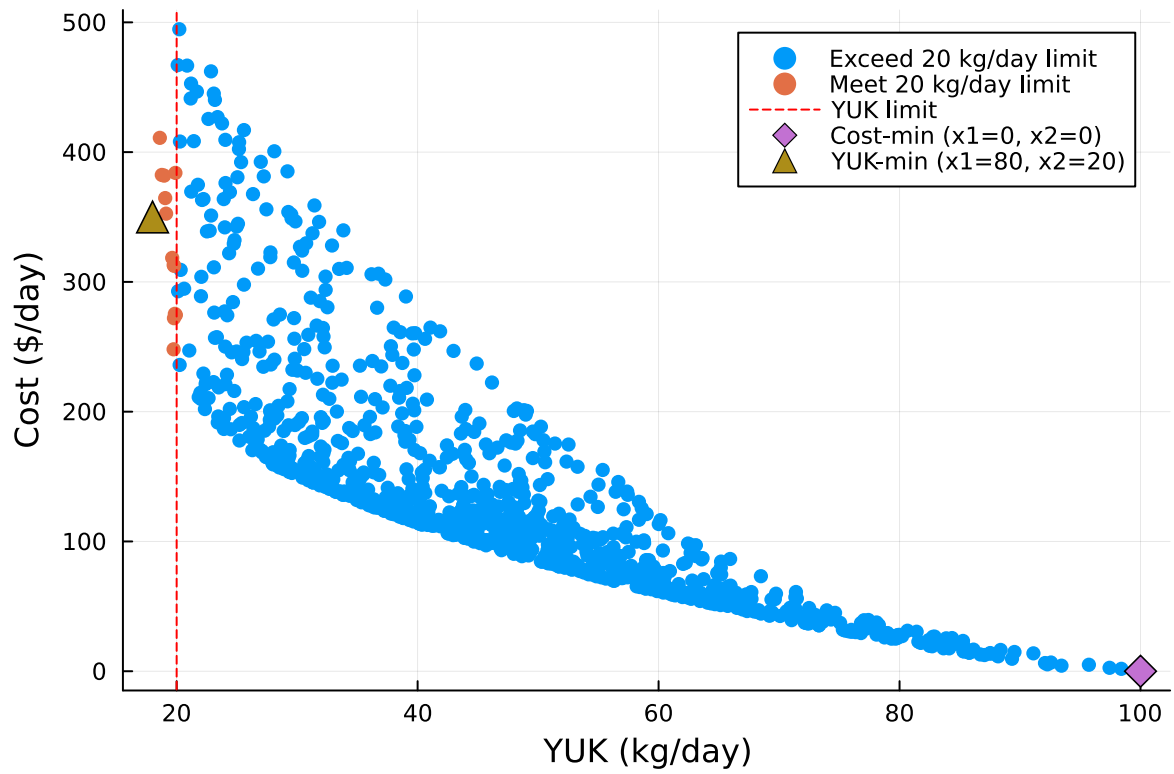
# Add points minimizing cost and discharge found in 2.4 to scatter plot
# point minimizing cost:
min_cost_d = 100
min_cost_c = 0

# point minimizing discharge:
min_discharge_d = 18
min_discharge_c = 350

# add points to plot
scatter!([min_cost_d], [min_cost_c],
        markershape=:diamond, ms=8, label="Cost-min (x1=0, x2=0)")
scatter!([min_discharge_d], [min_discharge_c],
        markershape=:utriangle, ms=8, label="YUK-min (x1=80, x2=20)")

```

```
display(current())
```



13×5 DataFrame

Row	x1 Float64	x2 Float64	x3 Float64	discharge Float64	cost Float64
1	74.572	23.3048	2.12323	19.7532	313.006
2	82.4794	16.2946	1.22596	19.0494	364.585
3	85.4656	12.5129	2.0215	19.8975	383.988
4	80.7119	17.9113	1.37688	19.1233	352.587
5	84.9461	14.0713	0.982667	18.9619	381.899
6	67.0319	31.5847	1.38339	19.7777	272.041
7	61.8042	38.0466	0.149248	19.7478	248.058
8	75.4528	22.5505	1.99664	19.6298	318.482
9	74.4768	23.3508	2.17236	19.794	312.366
10	67.6	30.6945	1.70551	19.9363	274.53
11	84.9703	14.2733	0.756385	18.7691	382.408
12	88.8396	10.9194	0.240975	18.6051	411.003

13 67.7337 30.6609 1.60541 19.8526 275.384

Make sure you comment your code appropriately to make it clear what is going on and why.

2.2 Solution: Explanation of model

This model takes in the parameters of x_1 , x_2 , and sets the total flow leaving the plant as $100 \text{ m}^3/\text{day}$. From these inputs, x_3 can be found by subtracting x_1 and x_2 from the total flow of $100 \text{ m}^3/\text{day}$. Then, the discharge and cost quantities can be found using these equations:

1. Discharge = $0.2 * x_1 + 0.005 * x_2^2 + x_3$
2. Cost = $(x_1 / 20) + 1.50 * x_2$

The discharge equation is explained in above in problem 2.1 and the cost equation is derived from the given cost of $x_1 / 20$ for the land treatment and \$1.50 per kg of wastewater being treated chemically from the x_2 stream.

Finally, the output of this model displays whether or not these inputs meet the requirement of having no more than 20 kg/day enter Pristine Brook. If it does meet this standard, the model returns true, but if it does not then false.

Problem 2.3

Use your function to experiment with 1,000 different combinations of wastewater discharge and treatment. You can do this with either a grid search or by sampling from a [Dirichlet distribution](#) (a $\text{Dirichlet}(1, n)$ distribution will generate uniformly-weighted n -dimensional vectors whose components add up to 1; see the [Distributions.jl documentation](#) for how to sample from probability distributions in Julia). Plot the results of these experiments. Do any satisfy the YUK effluent standard (plot this as well as a dashed red line). What was the cost of solutions satisfying the standard? What can you say about the tradeoff between treatment cost and YUK discharge? You don't have to find an "optimal" solution to this problem, but what do you think would be needed to find a better solution?

2.3 Solution:

I used a Dirichlet distribution as recommended to plot 1000 trials of possible wastewater allocations. I then put all the randomly generated allocations of x_1 , x_2 , and x_3 into a data frame so that every option could have its final YUK discharge and cost calculated and plotted.

Note: I used Julia documentation, StackOverflow, and ChatGPT to figure out how to code a scatter plot. ChatGPT was especially helpful with debugging. For example, my code was originally: $x_3 = 100 - x_1[s] - x_2[s]$, and I kept getting an error. ChatGPT informed me that I needed to add periods and change the line to: $x_3 = 100 .- x_1[s] .- x_2[s]$. This way Julia could execute vector/element-wise subtraction because $100 - \text{vector}$ is not defined.

Some trials, about 10% of them, do satisfy the YUK effluent standard. Most do not, however. The cost of solutions satisfying the standard range from roughly about \$250 - \$400. With respect to tradeoffs between treatment cost and YUK discharge, I can say that it is definitely costly to decrease the YUK concentration in the water that enters Pristine Brook. The graph clearly shows a downward trend where as YUK concentrations increase, cost decreases. So if Pristine Brook wants to be kept at safe levels of YUK, a few hundreds of dollars must be invested to make it happen. In order to find a better solution to this problem, it would be helpful to have a method of optimizing what allocations of wastewater would minimize cost but make sure to meet the 20 kg/day YUK concentration limit. This model could possibly consider the cost-effectiveness of land and chemical treatment to allocate flows to where they are removing YUK most cheaply and just until the 20 kg/day standard is met because any more than that although perhaps better for the Brook is not a necessary expense.

Problem 2.4

Find the strategies which minimize cost and YUK discharge (these will be different strategies) analytically and find the values of the objective metrics. Plot these values in the plot that you created for Problem 2.3. How do their values compare to the spread of values that you found in that problem? Would you select either of them (explain why or why not)?

2.4 Solution:

Minimizing cost:

Since treating water is what adds a cost, having all output from the plant be untreated would yield a cost of \$0.00, minimizing cost as much as possible. This is shown below:

$$x_1 = x_2 = 0 \text{ m}^3 / \text{day}, x_3 = 100 \text{ m}^3 / \text{day}$$

$$\text{Cost} = (x_1 / 20) + 1.50 * x_2 = \$0.00$$

However, YUK discharge = $0.2 * x_1 + 0.005 * x_2^2 + x_3 = 100 \text{ kg/day}$. This is above the 20 kg of YUK /day standard.

** To summarize: the minimum cost is when \$0.00 no water is treated ($x_1 = x_2 = 0 \text{ m}^3 / \text{day}$, $x_3 = 100 \text{ m}^3 / \text{day}$) and the discharge would be 100 kg/day, which is significantly above the 20 kg/day limit.**

Minimizing YUK:

To minimize YUK, all water should be treated in some way. This would mean that $x_3 = 0 \text{ m}^3 / \text{day}$ and we solve for x_1 and x_2 below:

$$\text{Since } x_3 = 0, x_2 = 100 - x_1$$

$$\text{YUK discharge} = 0.2 * x_1 + 0.005 * x_2^2 + x_3 = 0.2 * x_1 + 0.005 * (100 - x_1)^2 - 0$$

To find the minimum of the discharge equation, we use the derivative: $\text{discharge}' = 0.01 x_1 - 0.8$, and set it to zero. Finally, solving for x_1 yields: $x_1 = 80 \text{ m}^3 / \text{day}$.

If $x_1 = 80 \text{ m}^3 / \text{day}$, then $x_2 = 100 - x_1 = 20 \text{ m}^3 / \text{day}$. The final discharge amount at these wastewater allocations is 18 kg/day, which is in fact below the 20 kg/day limit. The cost of this process would be \$350/day.

To summarize: the minimum YUK discharge is 18 kg/day at a cost of \$350/day, when $x_1 = 80 \text{ m}^3 / \text{day}$, $x_2 = 20 \text{ m}^3 / \text{day}$, and $x_3 = 0 \text{ m}^3 / \text{day}$.

Compared to other plotted values, the minimum cost point is the furthest right and the minimum discharge point is the furthest left, showing two extreme possibilities in this scenario.

I would not chose either of these options. The minimum discharge option is pretty expensive and definitely not cost-efficient. The minimum cost option exceeds the limit of 20 kg of YUK /day by a lot which could be very harmful to the Pristine Brook and its ecosystem. I would try to look for an option optimizing the cost while meeting the 20 kg/day limit.

References.

List any external references consulted, including classmates.

For problem 1.3 and 1.4 I searched ChatGPT, StackOverflow, and Julia documentation to understand what MethodError means. StackOverflow and Julia documentation provided examples and fixes for the error. I asked ChatGPT exactly what the error means and it explained that incorrect arguments were given to a function.

For problem 1.4, I used Julia documentation to find the `copy()` method to make a new vector.

For problem 2, I used ChatGPT, StackOverflow, and Julia documentation in a few different ways and for different reasons. ChatGPT was really helpful for de-bugging especially when plotting in problem 2.3 . I explain more about how it helped within my solution explanation under 2.3 . I also checked my work with Holly Archer once we had both completed the entire problem set on our own. This was helpful to make sure that I had followed the directions correctly and got reasonable answers.

I checked my work with Holly Archer and Parker Catena separately, once each of us had fully completed the problem set on our own.